LEVEL UP ACADEMY

Class 9 | CBSE / GSEB | Science

Practice Pack: Motion

Chapter / Topic: Topic-based: use with whichever Class 9 book your students follow

Name: _____ Roll No.: _____ Date: _____

Marks obtained: _____

General instructions:

1. Level 1 checks the ideas, Level 2 is board-style numericals, and Level 3 is Foundation-level challenge.
2. In every numerical: write the given values, convert units (km/h → m/s: divide by 3.6), write the formula, substitute, and give the answer with its unit.
3. Equations of motion: $v = u + at$ · $s = ut + \frac{1}{2}at^2$ · $v^2 - u^2 = 2as$

LEVEL 1: CONCEPTS

Q1. Define displacement. Can the displacement of a body be zero when the distance it has covered is not zero? Give an example.

Q2. A body covers equal distances in equal intervals of time. What kind of motion is this?

Q3. What does the slope of a distance–time graph tell us?

Q4. What does the area under a velocity–time graph give?

Q5. Is uniform circular motion accelerated? Explain.

LEVEL 2: NUMERICALS

Q6. A car travels 150 km in 3 hours. Find its average speed in km/h and in m/s.

Q7. An athlete completes one round of a circular track of diameter 200 m in 40 s. What distance has he covered, and what is his displacement, at the end of 2 minutes 20 seconds? (Take π = 22/7.)

Q8. A bus slows down from 80 km/h to 60 km/h in 5 s. Find its acceleration.

Q9. A train starting from rest reaches a speed of 72 km/h in 5 minutes. Find its acceleration and the distance it travels in this time.

Q10. A car accelerates uniformly from 18 km/h to 36 km/h in 5 s. Find (i) the acceleration and (ii) the distance covered in that time.

Q11. A car moving at 20 m/s is brought to rest with a uniform retardation of 4 m/s^2. Find the time taken and the stopping distance.

Q12. A satellite moves in a circular orbit of radius 42,250 km and takes 24 hours to go around the Earth once. Find its speed.

LEVEL 3: CHALLENGE (FOUNDATION)

Q13. A car covers the first half of a journey at 40 km/h and the second half at 60 km/h. Find its average speed for the whole journey. (Hint: it is NOT 50 km/h.)

Q14. Car A starts from rest and accelerates at 2 m/s^2 for 5 s. Car B starts from rest and accelerates at 1 m/s^2 for 10 s. Which car ends with the greater speed, and which covers the greater distance?

Q15. A velocity–time graph is a straight line rising from 0 to 20 m/s in the first 4 s, then horizontal at 20 m/s for the next 6 s. Find the acceleration in the first 4 s and the total distance covered in 10 s.

~ Check every answer for units. That's where most marks go missing! ~